DAY 1 ROOFTOP SURVEILLANCE & BIOMETRIC REPORT

CHAIN OF CUSTODY
VERIFIED

Industrial Rooftop RO Water Plant • Camera IPC56 (3K HEVC $2880 imes 1620$)

Audit Batch ID: batch_20261023145845_884f05 | **Observation Cycle:** 2026-09-01 (Full 24h continuous coverage, 33/33 clips) |
Classification Engine: ArcFace 512-D + PostgreSQL pgvector HNSW

VIDEO COVERAGE	VACANT & LOCKED	PHYSICAL DWELL	FACE DETECTIONS
24h 00m	**22h 36m**	**1h 24m**	**181**
33 Files (100.0% Continuous)	94.15% Facility Unoccupied	19 Discrete Visit Sessions	7 Distinct Physical Clusters

TWO MULTIPORT VALVES	ULTIMATE POWER SWITCH	RO SKID CONTROLS ZONE
0 ACCESSES	**0 ACCESSES**	**5 SIGHTINGS**
100% Nominal / Untouched	100% Nominal / Untouched	Walkway & Rack Proximity

VERIFIED OBSERVATION: RO SKID CONTROLS & WALKWAY PRESENCE

SPATIAL AUDIT EVENT #SKID-01



Timestamp: 2026-09-01 09:14:42 AM | Frame Offset: 917.0s

Associated Subject: SUBJ-01 (Unassigned / Pending Enrollment)
Source Clip File: 2026-09-01_085926.mp4
Reference Evidence Clip: D:\facial\data\day1_incident_clips\incident_20260901_09144
Observed Location: RO Skid Controls Rack Envelope [945, 345, 1210, 510] (Face Box: [9 406])
Empirical Action: Subject positioned at the walkway adjacent to the RO skid controls rack. observed with the Two Multiport Valves or Power Switch.
Equipment Status: Two Multiport Valves: Untouched (0 accesses) | Ultimate Switch: Unt

Biometric Face Vectors & Unregistered Physical Clusters



SUBJ-01: Unregistered Subject #01

Name: Unassigned • Role: Unassigned

Status: Pending Identification / Enrollment
Observed across morning operating hours (07:37 - 09:18 AM). Interacted near RO Skid Controls rack at 09:14:42 AM. Zero contact with Two Multiport Valves or Power Switch.

Observed: 09:09:56 | Quality: 72.1% | Detections: 18 | Unit L2-Norm: 1.000000 | BBox: [1151, 424, 1214, 494]

ArcFace Unit Vector: [0.0027, 0.0554, -0.0089, 0.0525, -0.0101, 0.0155, -0.0280, 0.0159, ... [512 dims]]



SUBJ-02: Unregistered Subject #02

Name: Unassigned • Role: Unassigned

Status: Pending Identification / Enrollment
Observed on terrace walkway at 07:44 - 07:45 AM during morning hours. Zero equipment contact.

Observed: 07:45:33 | Quality: 80.0% | Detections: 24 | Unit L2-Norm: 1.000000 | BBox: [1342, 506, 1404, 581]

ArcFace Unit Vector: [0.0041, 0.0346, -0.0776, -0.1062, 0.0091, 0.0420, -0.0036, 0.0089, ... [512 dims]]



SUBJ-03: Unregistered Subject #03

Name: Unassigned • Role: Unassigned

Status: Pending Identification / Enrollment
Observed during dawn hours (06:23 - 06:36 AM) traversing terrace corridor.

Observed: 06:23:13 | Quality: 68.1% | Detections: 5 | Unit L2-Norm: 1.000000 | BBox: [1603, 388, 1665, 460]

ArcFace Unit Vector: [0.0078, -0.0045, -0.0361, 0.0229, 0.0216, 0.0135, 0.0882, -0.0447, ... [512 dims]]



SUBJ-04: Unregistered Subject #04

Name: Unassigned • Role: Unassigned

Status: Pending Identification / Enrollment
Observed during midday hours (11:18 - 12:04 PM) in corridor and near RO unit.

Observed: 11:18:10 | Quality: 79.5% | Detections: 51 | Unit L2-Norm: 1.000000 | BBox: [1697, 970, 1799, 1087]

ArcFace Unit Vector: [-0.0709, -0.0035, -0.0128, 0.0268, 0.0417, -0.0150, 0.0729, -0.1099, ... [512 dims]]



SUBJ-05: Unregistered Subject #05

Name: Unassigned • Role: Unassigned

Status: Pending Identification / Enrollment
Observed during afternoon hours (14:10 - 14:25 PM) traversing corridor and skid perimeter.

Observed: 14:22:45 | Quality: 82.3% | Detections: 22 | Unit L2-Norm: 1.000000 | BBox: [1628, 541, 1689, 615]

ArcFace Unit Vector: [-0.0387, -0.0346, -0.0117, 0.0004, 0.0379, 0.0326, -0.0692, -0.0170, ... [512 dims]]



SUBJ-06: Unregistered Subject #06

Name: Unassigned • Role: Unassigned

Status: Pending Identification / Enrollment

Observed during late afternoon hours (16:27 - 17:11 PM). Last detection recorded at 17:11:39 PM.

Observed: 16:32:52 | Quality: 77.4% | Detections: 37 | Unit L2-Norm: 1.000000 | BBox: [1520, 358, 1570, 422]

```
ArcFace Unit Vector: [-0.0425, 0.0384, 0.0067, 0.1084, -0.0670, 0.0546, 0.0203, 0.0036, ... [512 dims]]
```



SUBJ-07: Unregistered Subject #07

Name: Unassigned • Role: Unassigned

Status: Pending Identification / Enrollment

Observed during night hours (01:33 AM and 20:01 PM) on perimeter walkway.

Observed: 20:01:58 | Quality: 56.6% | Detections: 3 | Unit L2-Norm: 1.000000 | BBox: [1063, 844, 1176, 951]

```
ArcFace Unit Vector: [-0.1347, 0.0420, -0.1229, 0.0125, -0.0170, 0.0032, 0.0559, -0.0089, ... [512 dims]]
```

Chronological Audit Log of All 19 Operational Visit Sessions

#	Photo	Window	Dwell	Dets	Crew	Zone Breakdown & Telemetry	Valves	Switch	Qual
01		01:33:34 – 01:33:34	**0s**	1	1	CORRIDOR: 1	SAFE	SAFE	42.0%
02		06:23:10 – 06:23:13	**2s**	2	1	CORRIDOR: 2	SAFE	SAFE	68.1%
03		06:32:48 – 06:36:56	**4m 7s**	3	1	CORRIDOR: 3	SAFE	SAFE	57.4%
04		07:37:24 – 07:51:31	**14m 7s**	34	2	CORRIDOR: 25, RO_PLANT_AREA: 8, SKID_CONTROLS: 1	SAFE	SAFE	80.0%
05		08:01:40 – 08:02:37	**56s**	2	1	CORRIDOR: 2	SAFE	SAFE	70.4%
06		08:10:01 – 08:11:02	**1m 1s**	2	1	CORRIDOR: 2	SAFE	SAFE	61.5%
07		08:17:31 – 08:17:31	**0s**	1	1	CORRIDOR: 1	SAFE	SAFE	50.6%
08		08:59:31 – 09:18:50	**19m 18s**	22	2	CORRIDOR: 13, RO_PLANT_AREA: 5, SKID_CONTROLS: 4	SAFE	SAFE	72.1%
09		09:44:23 – 09:45:05	**42s**	2	1	RO_PLANT_AREA: 1, CORRIDOR: 1	SAFE	SAFE	55.7%
10		11:18:10 – 11:20:32	**2m 21s**	21	1	CORRIDOR: 15, RO_PLANT_AREA: 6	SAFE	SAFE	79.5%
11		11:26:09 – 11:28:57	**2m 47s**	5	1	CORRIDOR: 5	SAFE	SAFE	76.6%
12		11:59:13 – 12:04:37	**5m 24s**	25	3	CORRIDOR: 25	SAFE	SAFE	76.1%
13		14:10:06 – 14:25:03	**14m 57s**	21	2	CORRIDOR: 21	SAFE	SAFE	82.3%
14		14:30:46 – 14:30:46	**0s**	1	1	CORRIDOR: 1	SAFE	SAFE	60.2%
15		16:11:06 – 16:14:51	**3m 45s**	3	1	CORRIDOR: 3	SAFE	SAFE	66.3%
16		16:27:42 – 16:37:03	**9m 21s**	26	2	CORRIDOR: 24, RO_PLANT_AREA: 2	SAFE	SAFE	77.4%
17		17:00:56 – 17:00:56	**0s**	1	1	CORRIDOR: 1	SAFE	SAFE	51.2%
18		17:07:02 – 17:11:39	**4m 37s**	7	1	CORRIDOR: 7	SAFE	SAFE	76.3%
19		20:01:13 – 20:01:58	**44s**	2	1	CORRIDOR: 2	SAFE	SAFE	56.6%

Biometric Vector Pipeline & Database Indexing

ArcFace Hyperspherical Manifold Representation:
Each detected face crop is mapped via the ArcFace ResNet deep metric model into a 512-dimensional vector $\mathbf{v} \in \mathbb{R}^{512}$, normalized to the unit hypersphere $\mathbb{S}^{511}$ such that $\|\mathbf{v}\|_2 = 1.000000$. Pairwise cosine similarity is computed directly via the dot product $\text{Sim}(\mathbf{u}, \mathbf{v}) = \mathbf{u} \cdot \mathbf{v}$.

Supabase PostgreSQL Database Schema & Vector Index:
All detections and unit vectors are stored in PostgreSQL table `detections` with column `embedding_vector vector(512)`.
Vector similarity searches are accelerated using a **Hierarchical Navigable Small World (HNSW)** index:
`CREATE INDEX idx_detections_embedding_hnsw ON detections USING hnsw (embedding_vector vector_cosine_ops) WITH (m = 16, ef_construction = 64);`
Provides sub-1.5 ms query latency across 100,000+ vector records.

Production Roadmap for Days 2 through 30

Two-Tier Spatial Motion Optimization:
• **Tier 1 (Local 12-Core CPU at 400 FPS):** `scripts/fast_spatial_roi_scanner.py` scans remaining 990 files (Days 2 to 30) across folders `E:\RO WATER SEP\2..30`. Detects the ~5.8% active motion frames across RO skid and terrace corridor in **~5.5 hours** without internet uploads.
• **Tier 2 (Cloud GPU Batch ArcFace Extraction):** Processes only the extracted motion frames on JarvisLabs RTX GPU. Extracts all faces and vectors in **~1.5 hours** within ₹120 of the ₹200 budget.